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IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE/EIE PART III/IV: MEng, BEng and ACGI

MACHINE LEARNING

Friday, 5 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

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MACHINE LEARNING EXAM PAPER 2025

PLEASE ANSWER 3 OUT OF 4 QUESTIONS.

1. Linear Classification, Non-linear Feature Transformation & Perceptron Learning Algorithm

   a) Hoeffding's inequality can be written as:

$$P\left(|\hat{R}_n(h) - R(h)| > \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

   Answer the following:

   i) Starting from Hoeffding's inequality, state the generalization bound that holds with probability at least $1-\delta$ for a fixed hypothesis h. Then, Derive an expression for the minimum required sample size n in terms of the tolerance ε and confidence parameter δ for a fixed hypothesis h. **[15 marks]**

   ii) Using the generalization bound derived earlier, determine the minimum number of samples n required to ensure that

$$|\hat{R}_n(h) - R(h)| \leq 0.03$$

   with probability at least 99% for a fixed hypothesis h. **[15 marks]**

   b) Consider the following data samples along with their ground truth values y. Answer the following questions:

$\mathbf{x_1}$	$\mathbf{x_2}$	$\mathbf{y}$
-0.5	1	-1
0.5	-2	-1
-1	-2	+1
1	1	+1

   i) Plot the data on a Cartesian plane to visually demonstrate that the samples are non-linearly separable and cannot be classified with zero error using a linear classifier (e.g., a perceptron). **[5 marks]**

   ii) Apply the Pocket Algorithm using the **main Perceptron Learning Algorithm (PLA)** you learned in class for **5 weight updates** (not including the initialization).

At each update:

- Identify the **first misclassified sample** and use it to update the weights.
- After each update, **estimate the number of misclassified samples** on the entire dataset.

Finally, report the **pocket weight vector** $\mathbf{w}^{\text{poc}}$ and the corresponding **number of misclassified samples**. **[30 marks]**

iii) Apply the regression for classification approach by using closed form solution to find the weight vector. Based on the result, discuss whether this approach provides good initial weights for perceptron classifier.

[15 marks]

iv) Suggest a non-linear transformation $\Phi(\mathbf{x}) = (\mathbf{z}_1, \mathbf{z}_2)$ that will make the data linearly separable while maintaining the feature space dimension the same. Then, plot the transformed data on a Cartesian plane to visually demonstrate that the samples become linearly separable under the chosen transformation. **[20 marks]**

2. Multilayer Perceptron, Neural Network, Regularization, Validation and VC Dimension

a) The data in the table below cannot be classified with zero error using 1-D perceptron

x	y
1	+1
2	−1
4	+1

Suppose that we will use instead a network with two hidden perceptrons in parallel, followed by an output perceptron. The hidden perceptrons implement thresholds at $x = 1.5$ and $x = 3$:

$$z_1 = \text{sign}(x - 1.5), \qquad z_2 = \text{sign}(x - 3).$$

The output perceptron combines them as

$$h = \text{sign}(a_1 z_1 + a_2 z_2 + c).$$

i) Draw a schematic diagram of the two-layer perceptron described in the question. Label all nodes and connections clearly. **[7 marks]**

ii) For each data point $x = 1, 2, 4$, compute the hidden activations (z_1, z_2). **[8 marks]**

iii) Find values of a_1, a_2, c such that the network outputs the correct labels. Then plot h for $x \in [0, 5]$. **[15 marks]**

iv) In the $(z_1, z_2)-$plane plot the points representing the hidden-layer activations for the three input samples. Using your chosen weights, find the decision boundary equation and sketch the decision boundary of the output perceptron, indicating which side corresponds to each class. **[12 marks]**

v) Conclude that with this two-layer network, three points on the real line can be shattered. What does this imply about the VC dimension of this architecture compared to that of a single 1D perceptron? **[8 marks]**

b) We are interested in a text classification task. Specifically, given a sentence represented [1] by a feature vector $x \in \mathbb{R}^d$, its corresponding label y is $+1$ if the sentence is grammatically correct, and -1 otherwise. A colleague of yours

[1] For simplicity, we assume that sentences have been embedded in $\mathbb{R}^d$ through some feature transformation

told you about this very recent paper[2] to appear in *NeurIPS 2025*, where a kernel based framework was proposed that leverages a frozen pretrained large language models to generate support vectors and corresponding labels for any given classification task.

i) You used the proposed framework in the paper for the given task and were able to generate n support vectors and corresponding labels $(\hat{x}_1,\hat{y}_1),\ldots,(\hat{x}_n,\hat{y}_n),\in\mathbb{R}^d\times\{-1,+1\}$ for your task. However, the method only outputs noisy version of the true support vectors and corresponding labels $(x_1,y_1),\ldots,(x_n,y_n),\in\mathbb{R}^d\times\{-1,+1\}$. More precisely, for all $i\in[n]$,

$$\hat{x}_i\sim\mathcal{N}(x_t,\sigma^2 I_d)\quad\text{and}\quad\hat{y}_i=\begin{cases}y_t & \text{with probability}\quad 1-p(\hat{x}_i)\\ -y_t & \text{with probability}\quad p(\hat{x}_i)\end{cases}$$

where $\sigma>0$ and $p(x_i)\in(0,1)\backslash\{1/2\}$. Recall that x_i is a support vector with label y_i if and only if $y_i=w_\star^\top x_t+b_\star$. Show that

$$\hat{y}_i=\begin{bmatrix}w_\star\\ b_\star\end{bmatrix}^\top z_i+\varepsilon_i \tag{2.1}$$

where z_i can only be expressed in terms of $\hat{x}_i$ and $p(\hat{x}_i)$, and ε_i can only be expressed in terms of $y_i,\hat{y}_i,x_i,\hat{x}_i,p(\hat{x}_i),w_\star$ and satisfies $\mathbb{E}[\varepsilon_i|\hat{x}_i]=0$.

[15 marks]

Hint: Verify that $\mathbb{E}[\hat{y}_i|\hat{x}_i]=\alpha y_i$, where α only depends on $p(\hat{x}_i)$.

ii) Assume knowledge of $p(\hat{x}_i)$ for all i. Using $(z_1,\hat{y}_1),\ldots,(z_n,\hat{y}_n)$ formulate the least squares problem. Consider the following matrix notation

$$Z^\top=\begin{bmatrix}z_1 & \ldots & z_n\end{bmatrix}\in\mathbb{R}^{d+1\times n}\quad\text{and}\quad\hat{y}=\begin{bmatrix}\hat{y}_1\\ \vdots\\ \hat{y}_n\end{bmatrix}\in\mathbb{R}^n,$$

and write down the unconstrained solution the least squares problem. Denote such solution by $(\hat{w},\hat{b})$, and assuming that $(x_1,y_1),\ldots,(x_n,y_n)$ are independent verify that it is an unbiased solution of $w_\star,b_\star$. **[15 marks]**

iii) Your colleague believes that the $w_\star$ should be sparse. Suggest an approach to incorporate this belief into your minimization problem.

[2] **[Paper]** S. Ohsawa, Support Vector Generation: Kernelizing Zero-Shot Classifiers from Pre-Trained Language Models, NeurIPS 2025

Write the corresponding minimization problem with a regularization parameter λ. Your colleague believes that $\lambda \in \{1, 10, 100\}$, discuss what should be the best practice to find the best regularization parameter. **[12 marks]**

iv) In the two previous questions, you have been using a regression formulation to solve a classification task. Is this a good idea in general? Is there a better idea? **[8 marks]**

3. Support vector machines, kernel methods

a) You are given a dataset containing n input-label pairs, $(x_1, y_1), \ldots, (x_n, y_n) \in \mathscr{X} \times \{-1, 1\}$ where $\mathscr{X} \subseteq \mathbb{R}^d$. You goal is to learn a hypothesis $h : \mathscr{X} \mapsto \{-1, 1\}$ that fits the data from a class of hypothesis $\mathscr{H}$.

i) A colleague of yours suggests using an SVM to solve the problem. Define the hypothesis class $\mathscr{H}$ you use, and describe under what condition using SVMs is a good idea. **[10 marks]**

ii) For a given hypothesis in $h \in \mathscr{H}$, an input-label pair (x, y), the hinge loss between $h(x)$ and y is defined as $\ell_{\text{hinge}}(h(x), y) = \max(1 - yh(x), 0)$. Your colleague suggests to formulate the binary classification problem as follows:

$$\min_{h \in \mathscr{H}} \sum_{i=1}^{n} \ell_{\text{hinge}}(h(x_i), y_i) + R(h) \tag{3.1}$$

where $R(\cdot)$ is some regularization function. Show that with a proper choice of $R(\cdot)$, the resulting problem is equivalent to a soft-margin SVM problem formulation with parameter C. **[20 marks]**

iii) Regarding the hinge loss formulation:

A. Let $s = -yh(x)$. Plot the hinge loss as a function of s (i.e. plot the function $s \mapsto \ell_{\text{hinge}}(h(x), y)$), when s varies in $(-4, 4)$.

B. For a given (x, y), what is the gradient of $\ell_{\text{hinge}}(h(x), y)$ with respect to the parameters defining your hypothesis h.

C. What benefits do you see in using the formulation (3.1)?

[15 marks]

iv) Recall the soft-margin SVM formulation with parameter C:

$$\begin{aligned} &\min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \xi_i \\ &s.t. \quad y_i(x_i^\top w + b) \geq 1 - \xi_i \quad \text{and} \quad \xi_i \geq 0. \end{aligned}$$

Write down the Lagrangian of this problem and corresponding equations from the KKT condition and derive the corresponding Lagrange dual problem. **[15 marks]**

v) An oracle informed you that $x_{i_1}, \ldots, x_{i_{d+1}}$ are support vectors and that the matrix

$$\begin{bmatrix} x_{i_1} & \ldots & x_{i_{d+1}} \\ 1 & \ldots & 1 \end{bmatrix} \in \mathbb{R}^{d+1 \times d+1}$$

is full rank. Propose a solution to the problem that leverages this additional information, by explicitly expressing of the solution w and b. Give one benefit and one shortcoming to using this solution. **[24 marks]**

vi) You run the obtained solution on some test data, but you noticed that the test error is significantly larger than the training error. List two possible issues that could explain this poor performance, and propose a solution for each one of them. **[16 marks]**

4. Principal components analysis, Clustering, GMMs

a) You are given three 2D points:

$$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad x_3 = \begin{bmatrix} -2 \\ 0 \end{bmatrix}.$$

i) Compute the sample mean μ and the mean-centred dataset $\tilde{x}_i = x_i - \mu$.

[3 marks]

ii) Compute the empirical covariance matrix $S = \frac{1}{3}\sum_{i=1}^{3} \tilde{x}_i \tilde{x}_i^\top$.

[6 marks]

iii) Find eigenvalues/eigenvectors of S. Identify the first principal component u_1 (unit vector). **[5 marks]**

iv) Compute 1-D PCA coordinates $z_i = u_1^\top \tilde{x}_i$ and rank-1 reconstructions $\hat{x}_i = \mu + z_i u_1$. Briefly interpret $z_i u_1$. **[6 marks]**

v) If you kept only u_2, what fraction of total variance is captured? Comment. **[4 marks]**

vi) Why would we want to reduce the dimensionality of our data? Can you think of examples where this can be useful? **[4 marks]**

vii) Outline the PCA workflow for a K-dimensional representation and how to transform/reconstruct points. **[12 marks]**

b) Consider nine 1D points: $\{0, 1, 2, 6, 7, 8, 20, 21, 22\}$. Use Euclidean distance $d(x, x') = |x - x'|$ and merge one pair per step, breaking ties by lower x first. Compute the clusters, using single-linkage first, followed by max-linkage distance. **[20 marks]**

c) Consider k-means clustering on a set of datapoints.

i) Write the k-means objective with Euclidean distance. **[5 marks]**

ii) State the two Lloyd steps that monotonically decrease the objective.

[5 marks]

iii) Give the per-iteration time complexity in terms of numer of samples N,clusters k and number of features$D$. **[5 marks]**

iv) Explain why k-means fails on two concentric rings and give a remedy.

[5 marks]

d) Consider a K-component GMM in $\mathbb{R}^D$:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k), \quad \pi_k \geq 0, \quad \sum_k \pi_k = 1.$$

i) Write the complete-data log-likelihood using latent one-hot variables $z_{nk} \in \{0,1\}$ with $\sum_k z_{nk} = 1$. **[4 marks]**

ii) Derive the E-step updates $\gamma_{nk} = \mathbb{E}[z_{nk}]$ **[6 marks]**

iii) Derive the M-step updates for π_k, μ_k, Σ_k using $N_k = \sum_n \gamma_{nk}$. **[6 marks]**

iv) State when GMMs reduce to k-means and give one advantage of GMMs. **[4 marks]**